RAJALAKSHMI ENGINEERING COLLEGE

Sandhiya B 251801285-2025-AIDS S2

Started on	Monday, 24 August 2026, 9:12 PM
State	Finished
Completed on	Monday, 24 August 2026, 9:17 PM
Time taken	5 mins 20 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>
// Divide and Conquer: find index of first zero using binary search
int findFirstZero(int arr[], int low, int high, int m) {
    int result = m; // default: no zero found, means 0 count = 0

    while (low <= high) {
        int mid = low + (high - low) / 2;

        if (arr[mid] == 0) {
            result = mid;      // candidate for first zero
            high = mid - 1;    // search left half for an earlier zero
        } else {
            low = mid + 1;     // search right half
        }
    }

    return result;
}

int main() {
    int m;
    scanf("%d", &m);

    int *arr = (int*) malloc(m * sizeof(int));
    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    if (m == 0) {
        printf("0\n");
        return 0;
    }

    int firstZeroIndex = findFirstZero(arr, 0, m - 1, m);
    int zeroCount = m - firstZeroIndex;

    printf("%d\n", zeroCount);

    free(arr);
    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔

	Input	Expected	Got	
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Sandhiya B 251801285-2025-AIDS S2

Started on	Monday, 24 August 2026, 9:18 PM
State	Finished
Completed on	Monday, 24 August 2026, 9:18 PM
Time taken	46 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int findMajorityElement(int nums[], int n) {
    int candidate = 0;
    int count = 0;

    for (int i = 0; i < n; i++) {
        if (count == 0) {
            candidate = nums[i];
            count = 1;
        } else if (nums[i] == candidate) {
            count++;
        } else {
            count--;
        }
    }

    return candidate;
}

int main() {
    int n;
    scanf("%d", &n);

    int *nums = (int*) malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }

    int result = findMajorityElement(nums, n);

    printf("%d\n", result);
```

```
35      free(nums);
36      return 0;
37  }
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Sandhiya B 251801285-2025-AIDS S2

Started on	Monday, 24 August 2026, 9:19 PM
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Time taken	49 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Divide and Conquer: binary search for floor of x
int findFloor(int arr[], int low, int high, int x) {
    int result = -1; // if no element <= x exists

    while (low <= high) {
        int mid = low + (high - low) / 2;

        if (arr[mid] == x) {
            return arr[mid]; // exact match, floor is x itself
        } else if (arr[mid] < x) {
            result = arr[mid]; // candidate for floor
            low = mid + 1;     // search right half for a closer value
        } else {
            high = mid - 1;    // search left half
        }
    }

    return result;
}

int main() {
    int n;
    scanf("%d", &n);

    int *arr = (int*) malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    int result = findFloor(arr, 0, n - 1, x);

    printf("%d\n", result);

    free(arr);
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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S2

Started on	Monday, 24 August 2026, 9:20 PM
State	Finished
Completed on	Monday, 24 August 2026, 9:21 PM
Time taken	1 min 2 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Recursive Binary Search (Divide and Conquer) to find target in arr[low..high]
int binarySearch(int arr[], int low, int high, int target) {
    if (low > high) {
        return -1; // base case: not found
    }

    int mid = low + (high - low) / 2;

    if (arr[mid] == target) {
        return mid;
    } else if (arr[mid] < target) {
        return binarySearch(arr, mid + 1, high, target); // divide: search right half
    } else {
        return binarySearch(arr, low, mid - 1, target);  // divide: search left half
    }
}

// Recursive function to try each index for a valid pair
int findPair(int arr[], int n, int x, int i, int *elem1, int *elem2) {
    if (i >= n) {
        return 0; // base case: no pair found after checking all elements
    }

    int complement = x - arr[i];
    int foundIndex = binarySearch(arr, i + 1, n - 1, complement);

    if (foundIndex != -1) {
        *elem1 = arr[i];
        *elem2 = arr[foundIndex];
        return 1; // pair found
    }

    // Recurse to check the next index
    return findPair(arr, n, x, i + 1, elem1, elem2);
}

int main() {
    int n;
    scanf("%d", &n);

    int *arr = (int*) malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    int elem1, elem2;
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Sandhiya B 251801285-2025-AIDS

S2

Started on	Monday, 24 August 2026, 9:21 PM
State	Finished
Completed on	Monday, 24 August 2026, 9:22 PM
Time taken	1 min 6 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>
#include <stdlib.h>

void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

// Partition function: places pivot at correct position,
// smaller elements to left, larger to right
int partition(int arr[], int low, int high) {
    int pivot = arr[high]; // choosing last element as pivot
    int i = low - 1;       // index of smaller element

    for (int j = low; j < high; j++) {
        if (arr[j] <= pivot) {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i + 1], &arr[high]);
    return i + 1;
}

// Recursive Quick Sort (Divide and Conquer)
void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int pi = partition(arr, low, high); // divide: partition around pivot

        quickSort(arr, low, pi - 1);   // conquer: sort left part
        quickSort(arr, pi + 1, high);  // conquer: sort right part
    }
}

int main() {
    int n;
    scanf("%d", &n);

    int *arr = (int*) malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    quickSort(arr, 0, n - 1);

    for (int i = 0; i < n; i++) {
        printf("%d", arr[i]);
        if (i != n - 1) {
            printf(" ");
        }
    }
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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